Supplementary Material: Transformer based performance attribution for GPU kernels interference at PTX level

I. Pre-training Configurations

TABLE I
Hyperparameter configuration shared by all pre-training runs.

Hyperparameter	Value
Vocabulary size	8,000
Model dimension d_{model}	768
Number of layers	12
Number of attention heads	8
Feed-forward dimension d_{ff}	3,072
Max sequence length	2,048
Dropout	0.1
Batch size	16
Gradient accumulation steps	4
Effective batch size	64
Learning rate	5×10^{-5}
Weight decay	0.01
Max gradient norm	1.0
Warmup steps	1,000
Masking probability	0.15
Label smoothing	0.1
Number of epochs	6
Early stopping patience	3
Number of workers	10
Overlap	128
Seed	42

TABLE II
Number of trainable parameters and training duration per model variant.

Model	Number of Parameters	Training Duration
MLM	91,801,664	18 h 32 m 30 s
iMLM	91,801,664	18 h 34 m 14 s
iMLM + ITC	92,404,558	18 h 36 m 11 s

A. Generated Heatmaps Study on Examples

We study three progressively optimized implementations of 2D convolution with a 9×9 mask over a multi-channel image. We consider 2D convolution as an ideal first validation benchmark for this work because it is inherently L2-sensitive due to the overlapping input neighborhoods accessed by neighboring threads, its sensitivity is tunable by design through well-understood optimizations (shared memory staging, register tiling), and the expected direction of heatmap change at each optimization step is known a priori, providing a ground truth against which attribution output can be evaluated without a simulator. The three kernels span the canonical GPU optimization trajectory: `Convolution_2D_globalMemory` (1) assigns one output pixel per thread and fetches all input and mask data directly from global memory with no reuse mechanism, making every load a potential L2 miss under thrashing. `Convolution_2D_tiled_noSmem` (K2) introduces 2×2 register tiling, quadrupling the per-thread output footprint and unrolling the inner loop four times in PTX, but retaining exclusive global memory access for input data. `Convolution_2D_tiled_smem` (K3) adds cooperative shared memory staging: the thread block collectively loads the output tile and its 4-pixel halo into a shared buffer before accumulation, confining global memory traffic to a single load phase and fundamentally altering the L2 access pattern. Table VI reports the top 5 attributed instructions per method and phase for each kernel.

TABLE III
Hyperparameters shared across all fine-tuning phases.

Hyperparameter	Value
Batch size	32
Optimizer	AdamW
β_1, β_2	0.9, 0.999
ε	10^{-8}
Weight decay	0.01
Max gradient norm	1.0
Loss type	Huber
Huber δ	0.2
Scheduler	Cosine
Gradient accumulation steps	1
Early stopping patience	3
Fusion type	MLP
Dropout	0.1
Context dim	128
Feed-forward dim (fusion)	1,024
Time head hidden	128
Freeze encoder	No
Slowdown weight	1

TABLE IV
PER-PHASE FINE-TUNING CONFIGURATION AND TRAINING DURATION.

Hyperparams	Phase 1	Phase 2	Phase 3
PMU head outputs	none	9 (con.)	18 (con. + iso.)
PMU head hidden	64	128	128
Learning rate	1.4×10^{-4}	1×10^{-4}	1×10^{-4}
LR finetune	—	5×10^{-6}	5×10^{-6}
Warmup ratio	0.15	0.05	0.05
Min LR	10^{-6}	0	0
Num epochs	10	12	15
Early stopping	5×10^{-5}	10^{-4}	10^{-4}
Total parameters	100,188,289	101,075,338	101,962,387
Training duration	21 h 54 m 54 s	26 h 24 m 16 s	33 h 00 m 28 s

TABLE V
INSTRUCTION TYPE CLASSIFICATION CATEGORIES FOR ITC OBJECTIVE.

0	GLOBAL_LOAD	7	LOGIC_BITWISE
1	GLOBAL_STORE	8	COMPARISON
2	SHARED_LOAD	9	CONTROL_FLOW
3	SHARED_STORE	10	SYNC_BARRIER
4	LOCAL_LOAD	11	CONVERSION_MOVE
5	LOCAL_STORE	12	ATOMIC_REDUCE
6	ARITHMETIC	13	SPECIAL

B. Detailed Explaination : Evaluation of the Generated Heatmaps

1) Conv2d global memory:

a) Phase 1.: All three methods converge on `ld.global.f32` and `ld.global.u8` as the dominant contributors, which directly maps to the kernel's L2 vulnerability: the mask coefficients (`ld.global.f32`) are loaded 81 times per pixel per channel with no cross-thread reuse, and the pixel byte loads (`ld.global.u8`) generate sub-cacheline transactions that further stress L2 bandwidth. `add.s64 %rd29,%rd29,36` ranks consistently high across all three methods because it is the induction variable update for the mask base pointer: every mask load address in the inner loop is derived from it, making it the root of the entire load dependency chain—each of its 81 executions per pixel initiates a fresh sequence of cache-cold global loads. The `cvt.rn.f32.u16` conversions appear in SHAP's top 5 due to interaction terms: each immediately follows a `ld.global.u8` and absorbs part of its cache-miss cost in the marginal contribution accounting. The `.reg.b16 %rs<10>` directive appears in Attention's top 5 as the declaration that sizes the short-integer register file used by all subsequent `ld.global.u8` targets; the attention mechanism associates it with the volume of byte-granularity global traffic that follows. IG's completeness error of $0.09\,\%$ confirms the attribution is reliable.

b) Phase 2.: The addition of contention PMU supervision sharpens sensitivity to loop-stride structure: `add.s64 %rd29,%rd29,36` rises to rank 1 in IG and Attention, as the model learns that this pointer bump is the most direct predictor of L2 miss volume per inner loop execution. `st.global.u8 [%rd28],%r89` enters the top 5 in both IG and Attention: supervised on `con_l2_write_hit_rate`, the model correctly identifies that output write-back traffic also competes for L2 write bandwidth under contention. SHAP's ranking is stable with Phase 1, indicating the slowdown head's marginal contributions are not redistributed by the auxiliary PMU losses. IG completeness error remains $0.10\,\%$.

c) Phase 3.: With the full contention-vs-isolation contrast available, the model predicts an L2 read hit-rate drop of $\sim$0.44 (from 0.7918 in isolation to 0.3482 under contention), correctly identifying cache pollution as the primary slowdown mechanism. IG's top 5 shifts toward `or.pred` instructions: these implement the boundary guard that determines whether a filter position falls within the image or the zero-padded halo. Under L2 contention, boundary-divergent threads that evaluate the guard as false still consume a memory-transaction slot whose line has been evicted, and the model, now supervised on the isolation baseline, identifies the predicate logic gating those accesses as a meaningful contributor to the contention-isolation gap. `add.s64 %rd29,%rd29,36` remains present for the same structural reason as in prior phases. Attention preserves `add.s64` at rank 1 with `ld.global.f32` and `ld.global.u8` filling ranks 2–5, providing the clearest hardware-aligned picture across all phases. IG completeness error tightens to $0.03\,\%$. The convergence of all three methods on global load instructions and the mask-pointer induction update across all phases confirms that the kernel's lack of any data optimization mechanism makes it maximally sensitive to L2 thrashing, and that the model has correctly learned this structure from PTX alone.

2) Conv2d with register tiling:

a) Phase 1.: IG (completeness error $0.11\,\%$) recovers physically meaningful top instructions: `ld.global.f32 %f87,[%rd9+24]` (0.0137), `add.s64 %rd56,%rd56,36` (0.0113), and two `ld.global.u8` loads (0.0100, 0.0080), consistent with Kernel 1. `st.local.v4.f32 [%rd4],{%f37,%f37,%f37,%f37}` appears at rank 5 (0.0064): the compiler spills all four zero-initialized accumulators to local memory (DRAM) at the start of each channel iteration due to register pressure from the 2×2 tiling; its attribution reflects that this spill traffic adds latency on top of the global memory loads. Attention places `add.s64` at rank 1 (1.0064) and `st.local.v4.f32` at rank 4 (0.1943), consistent with IG. `cvt.s64.s32 %rd23,%r135` at rank 5 in Attention is the sign-extending cast promoting the 32-bit tile-pixel index to a 64-bit pointer offset; with four unrolled tile elements each requiring independent address computation, these casts are more numerous than in Kernel 1 and accumulate higher attention. SHAP

TABLE VI

TOP ATTRIBUTED INSTRUCTIONS PER ATTRIBUTION METHOD ACROSS PHASES, FOR EACH PROFILED KERNEL.

Kernel 1: Convolution_2D_globalMemory

P	SHAP	Integrated Gradients	Attention
1	6.350 ld.global.f32 %f32, [%rd29+12];	0.0193 ld.global.u8 %rs1, [%rd10];	1.0064 add.s64%rd29,%rd29, 36;
	5.983 ld.global.u8 %rs4, [%rd16];	0.0182 add.s64 %rd29,%rd29,36;	0.1741 ld.global.f32%f38, [%rd29+20];
	5.859 cvt.rn.f32.u16 %f28,%rs3;	0.0106 ld.global.u8 %rs4, [%rd16];	0.1075 .reg.b16%rs<10>;
	5.802 cvt.rn.f32.u16 %f31,%rs4;	0.0060 or.pred %p6,%p5,%p4;	0.0949 ld.global.u8%rs4, [%rd16];
	5.031 add.s32 %r2,%r1,-4;	0.0059 cvt.rn.f32.u16 %f46,%rs9;	0.0711 ld.global.f32%f44, [%rd29+28];
2	6.300 cvt.rn.f32.u16%f28,%rs3;	0.0442 add.s64 %rd29,%rd29,36;	1.0174 add.s64%rd29,%rd29, 36;
	5.418 ld.global.u8%rs4, [%rd16];	0.0184 or.pred %p6, %p5, %p4;	0.7553 ld.global.f32%f38, [%rd29+20];
	4.838 add.s32%r5,%r1, -1;	0.0167 or.b32 %r85, %r6, %r95;	0.2296 ld.global.u8%rs1, [%rd10];
	4.518 add.s32%r2,%r1, -4;	0.0163 st.global.u8 [%rd28], %r89;	0.2135 ld.global.u8%rs4, [%rd16];
	4.471 cvt.rn.f32.u16%f31,%rs4;	0.0142 ld.global.u8 %rs6, [%rd20];	0.1923 st.global.u8[%rd28],%r89;
3	5.600 cvt.rn.f32.u16%f28,%rs3;	0.0434 or.pred %p19,%p3,%p17;	1.0752 add.s64%rd29,%rd29, 36;
	4.900 add.s32%r2,%r1,-4;	0.0382 or.pred %p44,%p3,%p42;	0.7675 ld.global.f32%f38, [%rd29+20];
	4.195 add.s32%r5,%r1,-1;	0.0320 add.s64%rd29,%rd29, 36;	0.5169 ld.global.u8%rs4, [%rd16];
	4.017 ld.global.f32%f32, [%rd29+12];	0.0301 or.pred %p14,%p3,%p12;	0.3884 ld.global.f32%f35, [%rd29+16];
	3.431 ld.global.u8%rs4, [%rd16];	0.0178 or.pred %p29,%p3,%p27;	0.3114 ld.global.u8%rs1, [%rd10];

Kernel 2: Convolution_2D_globalMemory with Register Tiling

P	SHAP	Integrated Gradients	Attention
1	3.435 setp.gt.s32 %p20,%r1,%r58;	0.0137 ld.global.f32%f87, [%rd9+24];	1.0064 add.s64%rd56,%rd56, 36;
	2.806 cvt.s64.s32 %rd21,%r136;	0.0113 add.s64%rd56,%rd56, 36;	0.2188 ld.global.u8%rs1, [%rd16];
	1.603 .param.u32 <PARAM_5>;	0.0100 ld.global.u8%rs1, [%rd16];	0.1956 ld.global.f32%f75, [%rd9+8];
	- -	0.0080 ld.global.u8%rs17, [%rd51];	0.1943 st.local.v4.f32[%rd4], %f37,%f37,%f37,%f37;
	- -	0.0064 st.local.v4.f32[%rd4], %f37,%f37,%f37,%f37;	0.1730 cvt.s64.s32%rd23,%r135;
2	2.844 setp.gt.s32%p20,%r1,%r58;	0.022 add.s64%rd56,%rd56, 36;	1.0117 add.s64%rd56,%rd56, 36;
	2.751 cvt.s64.s32%rd21,%r136;	0.021 or.pred %p90,%p45,%p88;	0.7855 ld.global.f32%f87, [%rd9+24];
	1.572 .param.u32 <PARAM_5>;	0.012 ld.global.u8%rs18, [%rd53];	0.3502 st.local.v4.f32[%rd4], %f37,%f37,%f37,%f37;
	- -	0.011 st.local.f32[%rd6+4],%f110;	0.3459 ld.global.f32%f56, [%rd56+20];
	- -	0.010 ld.global.u8%rs1, [%rd16];	0.3061 st.global.u8[%rd55],%r123;
3	1.5971 cvt.s64.s32%rd21,%r136;	0.041 setp.lt.u32%p102,%r127, 20;	1.1284 ld.global.f32%f87, [%rd9+24];
	0.9127 .param.u32 <PARAM_5>;	0.017 or.pred %p88,%p51,%p86;	0.5701 add.s64%rd56,%rd56, 36;
	0.7000 ld.param.u64%rd12, [<PARAM_2>];	0.016 mad.lo.s32%r119,%r118,%r59,%r128;	0.4342 st.global.u8%r123;
	0.4374 mov.u32%r141, 0;	0.011 or.pred %p19,%p3,%p17;	0.0531 ld.global.u8%rs16, [%rd49];
	0.3359 setp.gt.s32%p20,%r1,%r58;	0.011 ld.global.u8%rs14, [%rd45];	0.0480 ld.global.u8%rs1, [%rd16];

Kernel 3: Convolution_2D_globalMemory with shared memory tiling optimization

P	SHAP	Integrated Gradients	Attention
1	9.331 fma.rn.f32%f39,%f38,%f37,%f36;	0.0127 st.shared.u8[%r43],%rs22;	1.1314 cvt.rn.f32.u16%f49,%rs19;
	7.870 fma.rn.f32%f22,%f21,%f20,%f19;	0.0125 ld.global.f32%f56, [%rd4+68];	0.7454 st.shared.u8[%r43],%rs22;
	6.441 ld.global.f32%f18, [%rd4+16];	0.0120 ld.shared.u8%rs8, [%r48+4];	0.5834 ld.global.f32%f56, [%rd4+68];
	6.260 cvt.rn.f32.u16%f34,%rs14;	0.0119 ld.shared.u8%rs4, [%r48];	0.5602 .extern.shared.align 16.b8 smem[];
	6.164 ld.shared.u8%rs11, [%r48+7];	0.0100 div.s32%r36,%r58,%r1;	0.4519 ld.shared.u8%rs14, [%r53+1];
2	6.354 ld.global.f32%f18, [%rd4+16];	0.0529 setp.lt.u32%p15,%r56, 20;	0.7305 ld.shared.u8%rs10, [%r48+6];
	6.110 cvt.rn.f32.u16%f34,%rs14;	0.0455 @%p15bra<BB_1>;	0.5413 st.shared.u8[%r43],%rs22;
	5.416 setp.ge.s32%p9,%r14,%r22;	0.0360 mul.lo.s32%r32,%r31,%r27;	0.3710 ld.shared.u8%rs11, [%r48+7];
	5.100 ld.shared.u8%rs10, [%r48+6];	0.0294 .reg.pred%p<16>;	0.3527 ld.shared.u8%rs6, [%r48+2];
	4.474 ld.shared.u8%rs17, [%r53+4];	0.0245 @%p13bra<BB_12>;	0.3503 ld.shared.u8%rs20, [%r53+7];
3	6.700 fma.rn.f32%f39,%f38,%f37,%f36;	0.0453 @%p15bra<BB_1>;	0.99438 ld.shared.u8%rs10, [%r48+6];
	6.260 cvt.rn.f32.u16%f34,%rs14;	0.0440 and.pred %p1,%p2,%p3;	0.99045 st.shared.u8[%r43],%rs22;
	6.234 ld.global.f32%f18, [%rd4+16];	0.0432 add.s32%r48,%r47,%r46;	0.77477 .reg.pred%p<16>;
	4.916 ld.shared.u8%rs10, [%r48+6];	0.0399 or.pred %p8,%p6,%p7;	0.77039 ld.global.f32%f15, [%rd4+12];
	4.742 setp.ge.s32%p9,%r14,%r22;	0.0334 cvt.rn.f32.u16%f55,%rs21;	0.52556 ld.shared.u8%rs20, [%r53+7];

however diverges: `setp.gt.s32 %p20,%r1,%r58` (3.435) and `cvt.s64.s32 %rd21,%r136` (2.806) dominate, with `.param.u32` ⟨PARAM_5⟩ third. This shift relative to Kernel 1 is not necessarily an attribution artefact: `setp.gt.s32` gates all four tile output writes simultaneously, giving it a genuine fan-out over the output computation that no single instruction in Kernel 1 possessed, so its high marginal contribution may be physically meaningful. `.param.u32` ⟨PARAM_5⟩ is the `channels` parameter declaration; SHAP captures that the channel loop trip count is the outermost multiplier of all memory traffic.

b) Phase 2.: IG completeness error tightens to $0.05\,\%$. `add.s64 %rd56,%rd56,36` rises to rank 1 (0.022), mirroring Kernel 1's Phase 2 behaviour under PMU supervision. `or.pred %p90,%p45,%p88` (0.021) enters rank 2: with four unrolled tile elements each requiring its own boundary guard, the OR-predicate combining per-dimension out-of-bounds flags is more numerous than in Kernel 1 and its attribution is amplified by PMU supervision. `st.local.f32 [%rd6+4],%f110` (0.011) is a scalar accumulator spill confirming register pressure remains a contributor across phases. Attention preserves `add.s64` at rank 1 (1.0117), promotes `st.local.v4.f32` to rank 3 (0.3502) and `st.global.u8` to rank 5 (0.3061), consistent with the model learning under `con_l2_write_hit_rate` supervision that store traffic competes for L2 write bandwidth. SHAP remains stable with Phase 1. The PMU predictions correctly capture `con_l2_write_hit_rate` ($\hat{z} = +0.376$, true $+0.380$) and `con_l1_hit_rate` ($\hat{z} = +0.474$, true $+0.464$), but `con_l2_read_hit_rate` is underestimated ($\hat{z} = +0.210$, true $+0.299$).

c) Phase 3.: The model's Phase 3 slowdown prediction ($2.11\times$) is the least accurate across all kernels and phases (true $2.28\times$), and attribution on a less confident prediction is inherently noisier; the following observations should be read with that caveat. IG completeness error is $0.04\,\%$. The top IG instruction shifts to `setp.lt.u32 %p102,%r127,20` (0.041): this predicate compares the loop counter against the profiling trip count of 20 (`ls_cat_iter_20`); its high attribution in Phase 3 suggests the model associates total memory-traffic repetition count with the contention-isolation gap, though whether this reflects a genuine learned relationship or a spurious correlation introduced by the lower prediction confidence cannot be determined from attribution alone. `mad.lo.s32 %r119,%r118,%r59,%r128` (0.016) computes a linearised pixel index for one of the four tile elements; its appearance is consistent with the model gaining sensitivity to address-generation arithmetic under isolation contrast. Attention shifts `ld.global.f32 %f87,[%rd9+24]` to rank 1 (1.1284), displacing `add.s64` (0.5701) to rank 2 for the first time across both kernels: with the full L2 delta signal available, the model attributes more directly to the cache-miss-generating load rather than its address-generating predecessor. SHAP remains dominated by `cvt.s64.s32` (1.597) and `.param.u32` (0.913). Overall, the attribution pattern for this kernel is less consistent across methods than Kernel 1, which could reflect the genuine structural ambiguity introduced by 4× loop unrolling—distributing the same logical operation across four equivalent instruction groups—the additional spill code introducing instruction classes less represented in training, or simply the lower prediction accuracy in Phase 3. The physically grounded instructions (`add.s64`, `ld.global`) remain identifiable in IG and Attention across all phases, confirming that the core L2 sensitivity signal is preserved even if SHAP's marginal contributions shift toward control-flow and indexing instructions.

3) Conv2d with tiled and shared memory:

a) Phase 1.: IG (completeness error $0.10\,\%$) places `st.shared.u8 [%r43],%rs22` (0.0127) at rank 1, followed by `ld.global.f32 %f56,[%rd4+68]` (0.0125) and two `ld.shared.u8` loads (0.0120, 0.0119). `st.shared.u8` is the cooperative tile-population write: every thread in the block issues one or more of these stores to stage the halo into shared memory; under L2 thrashing the global load that feeds it (`ld.global.f32`) suffers a cold miss, making the store the observable endpoint of the cache-pollution chain. `ld.shared.u8` reads are byte-granularity accesses into the staged tile during the convolution accumulation; their attribution reflects that even shared memory reads carry a latency cost when the tile was populated with cold-miss data. `div.s32 %r36,%r58,%r1` at rank 5 (0.0100) is the integer division computing each thread's linear tile index for cooperative loading; its attribution captures that the tile-index computation is the structural entry point to the entire cooperative load sequence. Attention places `cvt.rn.f32.u16 %f49,%rs19` at rank 1 (1.1314), `st.shared.u8` at rank 2 (0.7454), and `.extern.shared.align16.b8 smem[]` at rank 4 (0.5602). The shared memory declaration receiving high attention is significant: it is the single token that announces the existence and size of the shared memory buffer to the model, and the attention mechanism associates it strongly with the downstream `ld.shared` and `st.shared` traffic that defines this kernel's memory access pattern. SHAP is dominated by `fma.rn.f32` instructions (9.331, 7.870), a marked departure from Kernels 1 and 2 where load instructions led. This is physically meaningful: with 81 fused multiply-add operations per output pixel reading from low-latency shared memory, the accumulation arithmetic is no longer hidden behind memory latency as it was in the global-memory kernels—the FMAs become the compute bottleneck that the model identifies as most marginal to the slowdown prediction. The model predicted $2.10\times$ against a true $2.29\times$, the largest Phase 1 error across all kernels, suggesting the model finds this kernel's mixed global/shared memory profile harder to map to a slowdown scalar from PTX structure alone.

b) Phase 2.: IG completeness error is $0.14\,\%$, the highest across all kernels and phases, coinciding with the Phase 2 prediction being the most accurate ($2.28\times$ vs true $2.29\times$). The IG top 5 shifts entirely to control-flow instructions: `setp.lt.u32 %p15,%r56,20` (0.0529), `@%p15 bra ⟨BB_1⟩` (0.0455), `mul.lo.s32` (0.0360), `.reg.pred %p⟨16⟩` (0.0294), and `@%p13 bra ⟨BB_12⟩` (0.0245). `setp.lt.u32 %p15,%r56,20` and its branch implement the outer profiling loop trip-count check; their dominance under PMU supervision suggests the model learns that the number of full tile-load-and-compute cycles is the primary determinant of contention PMU accumulation. `.reg.pred %p⟨16⟩` is the predicate register declaration sizing the branch-predicate file; its attribution reflects the model associating the volume of conditional execution with the `con_stall_mio_throttle` and `con_stall_short_scoreboard` supervision signals. Attention shifts entirely to `ld.shared.u8` instructions (ranks 1, 3, 4, 5) with `st.shared.u8` at rank 2, consistent with the model under PMU supervision focusing on the shared memory traffic pattern that distinguishes this kernel from the global-memory kernels. SHAP promotes `ld.global.f32` (6.354) and `ld.shared.u8` (5.100, 4.474) over `fma.rn.f32`, reflecting that the auxiliary PMU heads redirect marginal contributions toward memory instructions. The PMU predictions correctly capture `con_l2_read_hit_rate` ($\hat{z} = +0.392$, true $+0.299$) but mispredict `con_stall_mio_throttle` in magnitude ($\hat{z} = +0.835$, true $+1.250$), consistent with MIO throttling being driven by the `st.shared` traffic pattern which is structurally novel relative to the training distribution.

c) Phase 3.: Phase 3 produces the highest completeness error across all kernels and phases (1.01 %), and the largest slowdown underestimation (2.03× vs true 2.29×). These two observations are coupled: a less confident prediction yields a smaller output gradient, compressing the attribution budget (+0.167868) relative to Phases 1 and 2 (+0.727, +0.776), which in turn makes the completeness error percentage larger even if the absolute residual is comparable. Attribution in this phase should therefore be interpreted with caution. IG top 5 is dominated by `@%p15 bra ⟨BB_1⟩` (0.0453), `and.pred %p1,%p2,%p3` (0.0440), `add.s32 %r48,%r47,%r46` (0.0432), and `or.pred %p8,%p6,%p7` (0.0399), with `cvt.rn.f32.u16 %f55,%rs21` (0.0334) at rank 5. The predicate and branch instructions implement the per-element boundary checks of the cooperative tile load; their dominance under the full contention-isolation contrast suggests the model associates divergent tile-loading behaviour with the L2 delta, since boundary threads issue fewer `st.shared` stores and leave parts of the halo unpopulated, altering the effective tile-load traffic seen by L2. `add.s32 %r48,%r47,%r46` computes the strided linear index for cooperative tile loading; its attribution is consistent with the model identifying tile-index arithmetic as the structural driver of how many global loads the cooperative phase issues. Attention converges on `ld.shared.u8` (0.994, 0.775) and `st.shared.u8` (0.990) at ranks 1–3, with `.reg.pred %p⟨16⟩` (0.775) and `ld.global.f32` (0.770) completing the top 5. This is the clearest hardware-aligned picture for this kernel: the shared memory load/store pair dominates because it is the instruction class unique to this kernel and most directly coupled to the tile-load phase that L2 thrashing degrades. SHAP is led by `fma.rn.f32` (6.700) and `ld.global.f32` (6.234), returning to the Phase 1 pattern and suggesting that under the compressed attribution budget the model reverts to the instructions with the highest absolute marginal contribution to the slowdown scalar. The iso PMU predictions show the model correctly captures `iso_l2_read_hit_rate` direction ($\hat{z} = +0.810$, true $+0.438$) but overestimates its magnitude, consistent with the model having seen fewer shared-memory kernels in the isolation regime during training. Overall, Kernel 3's attribution landscape is the most heterogeneous of the three: the coexistence of `ld.global`, `st.shared`, `ld.shared`, and `fma` in the top 5 across methods and phases faithfully reflects the kernel's two-phase memory access pattern—a global-to-shared staging step that is L2-sensitive, followed by a shared-memory accumulation step that is not—and the difficulty the model faces in disentangling their respective contributions to the observed slowdown is consistent with the higher prediction error and completeness error seen in this kernel.

$$C_{[i,j]} = \sum_{u=0}^{p}\sum_{v=0}^{q} A\left[i + u - \left\lfloor \frac{p}{2} \right\rfloor, j + v - \left\lfloor \frac{p}{2} \right\rfloor\right] \cdot B_{[u,v]} \quad (1)$$

where $A = \forall(a_{ij}) \in \mathbb{R}^{m\times n}$, $B = \forall(b_{ij}) \in \mathbb{R}^{p\times q}$, and $C = \forall(c_{ij}) \in \mathbb{R}^{m\times n}$. The superscript k identifies the convolution kernel associated with each compute kernel launch, with $k \in \{0, 1, \ldots, K-1\}$. Matrix A is considered persistent data.